

Anay Baid

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Technical Skills

Languages: C, C++, Python, JavaScript, TypeScript, Java, SQL, R, Kotlin, C#, HTML5, CSS3

Frameworks & Libraries: React.js, Node.js, Express, Redux, Spring Boot, Django, .NET, Flutter, TensorFlow, Pandas

DevOps & Tools: Docker, Kubernetes, AWS, Azure, Linux, Git, CI/CD, Selenium, JUnit, MongoDB, Jira, Postman

Professional Work Experience

Software & Network Optimization Engineer — Purolator Inc 🌐

Spring 2026

- Cut planning turnaround 10-20× and diagnostics 40% by owning backend development of **Python/PowerShell ETL pipelines**, **Airflow DAGs**, and **Flask REST APIs** processing **100K+** forecast records with **SQL-backed state**, validation, and reconciliation.
- Containerized forecasting workloads with **Docker**, using **Amazon S3 + Parquet** for forecast artifacts and **Amazon SageMaker** for managed inference; deployed via **ECR/ECS/Fargate**.
- Provisioned the **Airflow** → **S3** → **SageMaker** → **ECS/Fargate** architecture with **Terraform**, IAM-scoped workloads, private networking, **Secrets Manager**, and **KMS**.
- Developed **Python constraint-aware allocation algorithms** preserving network volume targets, exposing capacity bottlenecks and supporting **10-15% projected throughput gains**.

Software Engineer — Iotum Global Holdings Inc 🌐

May 2025 - August 2025

- Engineered full-stack **React reference applications** to demonstrate complex **SDK integrations**, increasing external developer adoption & reducing client implementation time by **30%**.
- Architected scalable test automation frameworks using **JavaScript**, **Selenium**, and **Mocha**, increasing code coverage by **85%** & enabling faster **CI/CD release cycles**.
- Diagnosed & resolved critical **user-facing latency issues**, improving application stability under high concurrency for **10,000+** users.

DevOps Engineer — Nationwide Appraisals 🌐

September 2024 - December 2024

- Modernized legacy corporate infrastructure by containerizing services with **Docker**, optimizing **TCP protocols & Layer 3 network configurations** to reduce system latency.
- Secured sensitive financial data pipelines using **SQL Workbench** & encrypted **SSH tunneling**, ensuring compliance & integrity across **AWS S3** and hybrid cloud environments.
- Orchestrated reproducible development environments for mobile testing by integrating **Java 8 JDK** with **Android Emulators**, accelerating debugging.
- Implemented automated device compliance via **Azure Cloud** & **Intune MDM**, reducing manual provisioning overhead by **40%**.

Data Engineer — Cineplex Entertainment LP 🌐

January 2024 - April 2024

- Engineered real-time **ETL pipelines** for the **Cineplex Homepage Dashboard**, ensuring high-availability box office reporting via automated data validation scripts.
- Optimized **Machine Learning algorithms** for show-time scheduling, improving prediction accuracy by **15%** via rigorous model evaluation & input data cleansing.
- Built automated data governance tools using **Python** and **SQL schema comparison**, validating **10TB+** of **Azure data** to ensure **100% integrity**.
- Led data quality assurance across hybrid **Azure** & **On-Prem** systems, identifying & resolving **200+** anomalies with automated **JUnit** and **Selenium** test suites.

Hackathons & Project Experience

StarRez Enterprise Automation Suite 🌐 | JavaScript, DOM Injection, Workflow Automation

February 2026

- Architected a production-grade automation extension for **100+** **Front Desk staff**, reducing manual logging by **80%** via automated DOM data extraction & processing pipelines.
- Designed a scalable, modular workflow engine in **JavaScript**, enabling campus-wide deployment approved by **University IST Management** 🌐; supports high-volume student record processing with zero manual intervention.

CI/CD Build Automation Tool 🌐 | Python, GitHub Actions API, SMTP, REST

UOttHack 5

- Won **Best Use of GitHub** prize for developing a dispatch tool that triggers targeted build pipelines via the **GitHub API**, streamlining cross-environment deployments.
- Implemented secure **SMTP notifications** & environment validation in **Python**, enabling auditable builds.

Education

University of Waterloo — Honours Computer Science 🌐

Exp. April 2027

- Achieved **100%** in **CS 445 (Software Requirements Specification)** & **CS 446 (Software Design and Architectures)** 🌐; recognized by the Director of Software Engineering.
- Delivered academic support for **CS 136/136L (Algorithm Design & Data Abstraction)** via **CS & Math Tutoring** 🌐; addressed technical queries & coordinated review sessions.